

DRIVECARE PRO

Automotive Mobile App for Vehicle Owner Self-Service

**DATABASE SYSTEMS ENGINEERING &
DISTRIBUTED BACKEND DEVELOPMENT**

TEAM 18

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Vehicle Self-Service → Smart Booking → Digital Records → Assistance

Problem Statement

Why traditional vehicle service management needs a digital platform

01

Service Booking

Calls and physical visits take time.

02

Records

History, invoices and warranty details are difficult to manage.

03

Emergency Gaps

Roadside assistance needs a faster digital channel.

04

Limited Updates

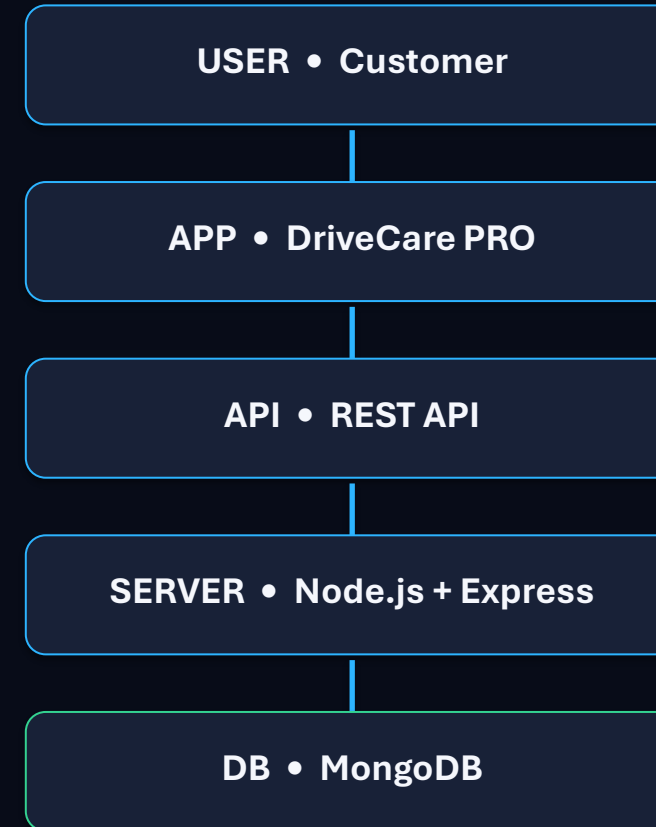
Customers need timely service and appointment information.

Proposed Solution

DriveCare PRO brings vehicle-owner tasks into one platform

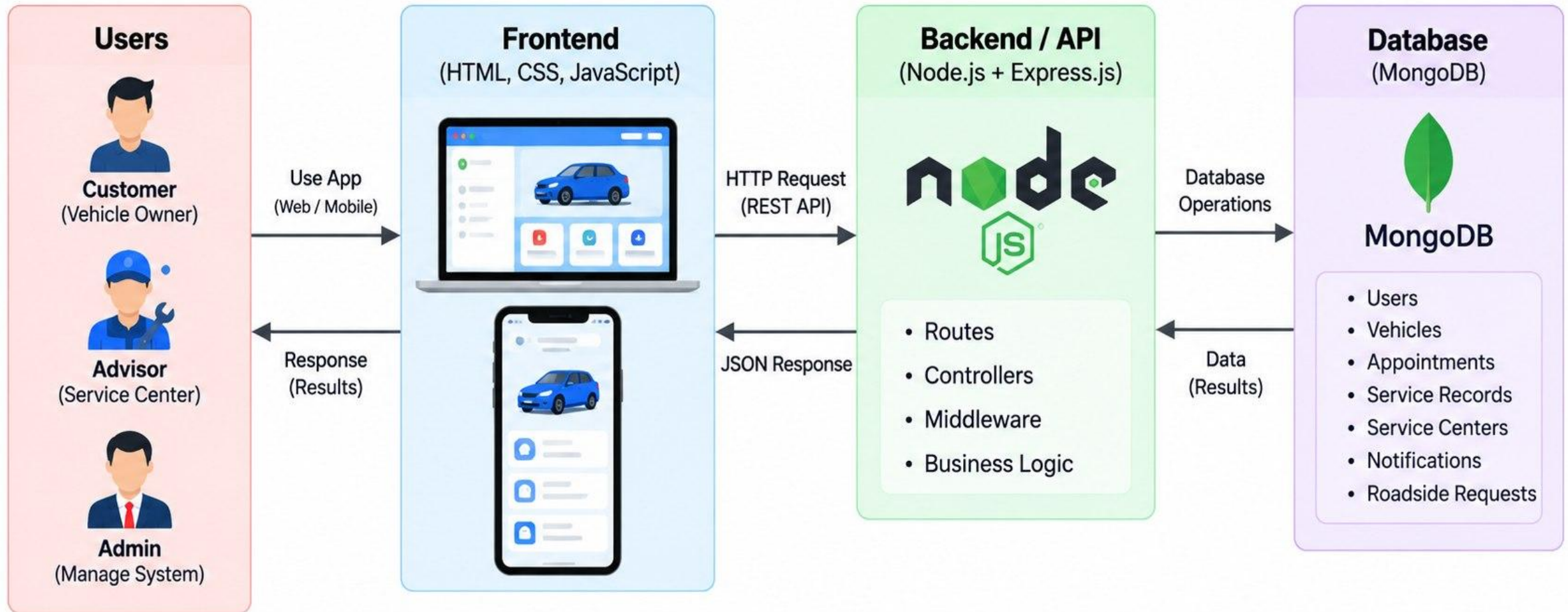
- User registration & login
- Vehicle management
- Service appointment booking
- Digital service records
- Service centers
- Roadside assistance
- Notifications & profiles
- Advisor / workshop functions

ONE APP



System Architecture

DriveCare PRO – Automotive Mobile App for Vehicle Owner Self-Service



Authentication & User Management

Secure access using credentials, JWT and protected routes

- Registration with user details
- Login using email + password
- Password verification
- JWT token generation
- Protected API routes
- Customer / Advisor / Admin roles

LOGIN FLOW:



Vehicle & Appointment Management

Create, schedule and track service appointments

- Select vehicle and service center
- Choose service type
- Select date and time slot
- Pickup / drop requirements
- Estimated cost & customer notes
- Track booking status

STATUS

Pending



Confirmed



In-Progress



Completed

Database Design

MongoDB + Mongoose models used by the application

User

Vehicle

Appointment

ServiceRecord

Notification

ServiceCenter

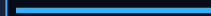
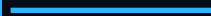
RoadsideRequest

REFERENCES

USER

VEHICLE

APPOINTMENT



Frontend & User Interface

Dark, modern and responsive DriveCare PRO experience

→ Dashboard

→ Vehicles

→ Appointments

→ Service Records

→ Roadside Assistance

→ Service Centers

→ Notifications

→ Profile / Advisor

THEME OPTIONS

Blue

Emerald

Purple

Sunset

Security, Notifications & Emergency Services

Supporting services that keep the platform connected and protected

- JWT authentication
- Authorization headers
- Protected routes
- Password verification
- Appointment notifications
- Roadside assistance requests
- Service-center information

SECURITY FLOW



PROTECTED REQUEST

Only authenticated users can access protected application functions.

Conclusion & Future Scope

From traditional servicing to a centralized digital experience

CURRENT SYSTEM

- ✓ Authentication
- ✓ Vehicle management
- ✓ Appointment booking
- ✓ Service records
- ✓ Service centers
- ✓ Notifications
- ✓ Roadside assistance

FUTURE SCOPE

- AI-based maintenance prediction
- Live roadside assistance tracking
- Online payment integration
- Mobile application using Flutter
- Advanced analytics

DRIVECARE PRO

A secure full-stack platform connecting vehicle owners, service centers and backend services.